

KuantorFlow — Weekly Development Report

Period: 25 July – 2 August 2026 · Repositories: kuantorflow, ai_agent, kuantorflow_automation

Executive Summary

This period delivered the **users and ownership cluster end to end** — from "the app has no idea who you are" to signed-in identities, per-card ownership, admin powers, account deletion, blocking, and a private-cards mode. **42 pull requests merged** across the three repositories and the regression suite grew from **114 to 464 passing offline tests**, a fourfold increase driven by the parallel-test convention.

It also included the project's **first production incident**, on 2 August: card saving returned errors for real users because a deploy silently skipped a schema change. The fix was not only the missing column but the deploy step itself, which is now a re-runnable, self-reporting command.

Repository	Role this period	Merged PRs
kuantorflow	The users/ownership cluster, action logs, notes formats, the schema deploy step	19
kuantorflow_automation	A parallel test PR for every feature, 13 verification reports, suite → 464	20
ai_agent	Away-aware recaps, and Mykola's second tool	3
Total		42

24 issues closed — 19 completed, four closed as *not planned* after being specified and thought through (soft delete #159, the Recycle Bin #160, delete-requests-to-admin #139, and the Allura welcome font #104, which was implemented and then declined on the look), and one as a duplicate. The reasoning was recorded on each rather than the issue being left open indefinitely.

Per the standing request, commit authorship was checked across all three repositories for the period: all 90 commits were authored by Kuantor (Claude as co-author). No GitHub Copilot commits were found in this range.

Completed Work by Theme

1. Identity and ownership — the cluster, complete

The site went from anonymous-only to a real multi-user model, in dependency order:

- **A `users` table** (#148, PR #166) — keyed on Google's OIDC sub, not the email. A user who changes their Gmail address keeps the same row and the same cards, instead of appearing as

a second person.

- **Every card records who added it** (#89, PR #169) — `added_by_user_id`, taken from the server-side session only, never from posted form data.
- **Delete only your own cards** (#162, PR #173) — enforced in the route as one conditional DELETE; the admin may delete any.
- **An admin identity** (#158, PR #172) — `ADMIN_EMAILS` in the server `.env`, requiring Google's `email_verified` claim. Configuration rather than a database column, deliberately: the admin's job is to block accounts, so admin-ness must not live in the table the blocking flow edits.
- **Delete my account** (#165, PR #179) — self-service, with an explicit keep-or-delete choice for the user's cards, plus an admin command-line script for accounts that cannot remove themselves.
- **Only an account may change the database** (#125, PR #182) — anonymous visitors can read everything and write nothing; every save path refuses in the route, not by hiding a button.
- **Blocking an account** (#126, PR #183) — a blocked learner keeps reading but cannot write or use Mykola, and is shown the admin's address so they can ask for access back.
- **Individual cards mode** (#127, PR #187) — an opt-in setting that hides everyone else's cards from the topic list, deck and quiz.
- **#93, the multi-user umbrella**, was re-read against what had shipped and **closed as delivered** by the four issues above.

Two supporting fixes landed alongside: per-user settings and chat logs were re-keyed on the user id rather than the email prefix (#174, PR #175), and the chat widget stopped replaying a previous user's conversation after a sign-out (#170, PR #171).

2. The deploy step that skipped schema changes

On 2 August, saving a card began returning errors in production. The cause was not the code that shipped but the **deploy instructions**: `schema.sql` creates tables with `CREATE TABLE IF NOT EXISTS`, which does nothing to a table that already exists, and the `ALTER` statements that would add a new column lived in that file only as **comments**. Re-applying the file reported success either way.

#180 (PR #181) replaced it with `apply_schema.py`: one command that creates missing tables *and* applies pending column changes, skips whatever is already present, prints every step it applied and every one it skipped, and exits non-zero on failure. The migrations are now real statements in an ordered list rather than prose. Its first production run was a clean no-op, which was itself the proof that the idempotency works — and #126's two new columns were the first migration written under the new scheme.

3. Mykola

- **Away-aware recaps** (`ai_agent`#54, PRs #60 and `kquantorflow`#156) — after a configurable break, Mykola reviews the last exchanges and opens a fresh conversation acknowledging how long the learner was away.
- **Anonymous message limits** (#164, PR #168) — a per-session nudge and a per-day ceiling that actually bounds the API bill, counted atomically so concurrent workers cannot both slip past the last message.

- **Mykola remembers what to call you** (ai_agent#62, PRs ai_agent#65 + kuantorflow#188) — a second agent tool. Saying "*Anna Maria is a mouthful, call me Ann*" stores the preference and he uses it from the next message on; "*go back to my proper name*" clears it. The tool is defined in the agent, but the callable that touches the database is injected by KuantorFlow, so the agent still never talks to MySQL.

4. Content and logging (earlier in the period)

- **Notes uploads accept `.txt` and `.docx`** as well as `.mht` (#137, PR #154) — one parser per format feeding a shared state machine.
- **Action logs** (#30, PR #155) — a plain-text, greppable trail of cards created, skipped, deleted and refused; which dictionaries were used and how long they took; and every notes upload. Logging never breaks a user action.
- **One close cross for the review popup** (#146, PR #153) and the `.mht` results-window behaviour (#45).

5. Process

- **`CLAUDE.md` in all three repositories** (#152, PRs #151/#59/#22) — the guidance a new contributor or agent needs: architecture, conventions, and the traps.
- **13 verification reports** committed this period, one per significant change, each recording what was checked and how.
- The suite reached **464 offline tests** running in about six seconds, plus opt-in round-trip tests against a real local database.

Technical Highlights

- **Key on what cannot change.** The users table is keyed on Google's sub claim rather than the email, because an email is user-editable and a key must not be. The same reasoning made `blocked_at` a nullable timestamp rather than a boolean: it answers *is this blocked* and *since when* in one column.
- **Enforce in the route, not the template.** Greying a control is presentation, and a hand-made request goes straight past it. Every rule this period — delete-your-own, sign-in-to-write, blocked accounts — is enforced where the request is handled, with the UI merely reflecting it.
- **One funnel, so a rule cannot be forgotten.** All card writes already went through a single function; the anonymous-write rule was enforced *there*, so a future save path that forgets to ask fails loudly instead of quietly writing.
- **`NULL` is not a value.** `added_by_user_id = 5` never matches an unowned card, which is exactly right for "only my cards" — and exactly wrong if "no filter" is expressed as an owner of `None`, which would hide the entire deck. The individual-cards filter is built around that distinction.
- **Read live what must be current.** A block is read from the database on each signed-in request rather than stamped into the session, because a session cookie lasts thirty days and a block must take effect on the next request.
- **Tools the model can reach for.** Mykola's second tool follows the first one's shape exactly: the agent defines it and returns a status the model relays in character, while the host injects the callable that touches the database — and refuses by raising, which becomes an honest in-character apology rather than a crash.

Lessons Learned

1. **A comment is not a deploy step.** The outage was possible because a required ALTER lived inside a file the deploy already ran, so following the instructions faithfully still skipped it. Anything a deploy must do belongs in something that *runs*, reports what it did, and fails loudly.
 2. **A stored shape that gains a field needs a decision about the old shape.** This bit three times in one session — a session dict gaining an id, widget state gaining an owner, a session gaining a verification claim. In each case the old shape read as a *valid* new one rather than as missing, so nothing failed loudly.
 3. **Tightening a rule invalidates the tests that codified the old freedom.** Denying anonymous writes broke fifteen tests; most needed only a fixture swap, but four had to be rewritten because their subject — a card saved with no owner — is behaviour that no longer exists. That distinction is worth budgeting for.
 4. **Verify the thing that cannot be unit-tested.** Whether the model actually *reaches for* a new tool is not something a stubbed test can answer, so the preferred-name feature was checked against the live model in four real conversations, including both refusal paths.
 5. **An audit trail should count events, not attempts.** Unblocking an account twice initially wrote two log lines though the second changed nothing — found while verifying, not while testing.
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Plans for Next Week

Taken from the KuantorFlow Improvements GitHub Projects board ([*dashboard*](<https://github.com/users/Kuantor/projects/2>)), in board-column order.

Highest Prio

Repo	Issue	Title
kuantorflow	#176	Edit a saved card: every field, including the word
kuantorflow	#184	Create new UI page
kuantorflow	#185	Images for words and topics
kuantorflow	#163	Do not write chat logs for anonymous visitors
kuantorflow	#144	Add Ukrainian/Russian translations after parsing MHT files
kuantorflow	#100	Quiz: treat perfective/imperfective answers as equal (AI)
kuantorflow	#99	Show a message about API credits when needed
ai_agent	#63	Move Mykola from Claude Opus 4.8 to Opus 5
ai_agent	#19	Teach Mykola to complement and extend card meanings
ai_agent	#61	Show message times in the chat

#176 is the natural next step: it gives the action log's edit entry its first call site, and it is the last unbuilt half of #125 — there is currently no edit route for the "change" rule to protect.

Nice To Have

Repo	Issue	Title
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kuantorflow	#177	Move a card to a different topic
kuantorflow	#178	Rename a topic (bulk) — admin-only
kuantorflow	#136	Images on flashcards (files on disk, path in MySQL)
kuantorflow	#92 / #94	Spaced-repetition review mode (SM-2)
kuantorflow	#22	Redesign website for a modern look
kuantorflow	#161	Make logs more verbose
kuantorflow	#85 / #88 / #90	Full-screen widget / backup docs / logs disclosure
ai_agent	#64	Cache Mykola's system prompt (prompt caching)
ai_agent	#56 / #57 / #58	Bigger avatar / add-card confirmation / save conversation
kuantorflow_automation	#4	Automated tests for the Mykola AI agent

ai_agent [#51](https://github.com/Kuantor/ai_agent/issues/51) (user-preferred nicknames) is now largely delivered by #62 and worth re-reading before any work starts on it.

In Progress — the word-games cluster

Repo	Issue	Title
kuantorflow	#130	Levenshtein multiple-choice translation game
kuantorflow	#131	Keyboard-adjacency typo distractors (MCQ)
kuantorflow	#132	"Is this a real word?" (n-gram pseudowords)
kuantorflow	#133	Typoglycemia scrambling mode
kuantorflow	#110	Investigation: Learner's vs Collegiate dictionary (+ M-W API)

Todo (selected)

Repo	Issue	Title
kuantorflow	#186	Duplicate prevention and individual-cards mode disagree about what exists
kuantorflow	#74	Dark theme for the website
kuantorflow	#25	Export/import cards (local ↔ remote DB)
kuantorflow	#120	Pin the Settings close button while the dialog scrolls
kuantorflow	#140	[Optional] Upload notes: support .pdf (text-layer)
kuantorflow_automation	#14	Refresh the test catalog — it lists 94 tests against 464 today

Outstanding operational item

#125, #126, #127 and ai_agent#62 are merged but not yet confirmed deployed. The deploy needs a `git pull` of *both* `kuantorflow` and `ai_agent`, then `apply_schema.py` — which is **not** a no-op this time, as it adds #126's two columns — followed by a web-app reload.

Report generated 2 August 2026 from GitHub pull-request, issue, project-board and commit data across the three repositories. Merged-PR window: 25 July 2026 (kuantorflow#151) through 2 August 2026 (kuantorflow#188, kuantorflow_automation#43, ai_agent#65).